$K > \sqrt{P}$

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☆ Numbers

1] Prime no:
- start from 2
- 1 is not prime

Q To test is ~~no even~~ given no P is prime

- $K > \sqrt{P}$
- write all prime no is less or equal to K
- check that prime no divide p or not
- IF not, then p is is prime otherwise not prime

→ $14 > \sqrt{191}$ (take $(14)^2$ becouse it is greater the 191)

prime less then 14 are

2, 3, 5, 7, 11, 13 ~~14 ≠ 0~~

No no is divide 191

191 is prime.

Q P=221

$K > \sqrt{P}$

$15 > \sqrt{221}$

prime less then is are

2, 3, 5, 7, 11, 13.

No no is divide 191

191 is not prime

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2] composite no

is writen a multiplication of prime no.

$14 = 2 \times 7$

$50 = 2 \times 5 \times 5$

$8 = 2 \times 2 \times 2$

Test by divisiblity

1] Divisible by 2

- checks only last digite is divisible by 2

Ex. 23456

6 is divisible by 2

2] Divisible by 3

- The sum all digite of given no is divisible by 3

Ex. 587421

$= 27/3$

it is divisible by 3

3] Divisible by 4

• ~~check last 2 di~~
• check ~~sum~~ of last 2 digit is divisible by 4.

Ex.

6 3 8 2 1 4

• 14 is not divisible by 4

5] Divisible by 5

• check 0 or 5 is present in last digit or not

5] Divisible by 6

• ~~2 & 3 Divis~~
• both 2 & 3 Divisible condition satisfy then it is Divisible by 6

Ex.

1] 9 7 4 5 6 2 = 33

It is Divisible by 6

2] 9 7 5 4 1 6 = 22

It is not Divisible by 6

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6] Divisible by 8

- ~~check last 3 digit~~ The no from of last 3 digit is divisible by 8

Ex. 6754120

120 is divisible by 8

7] Divisible by 9

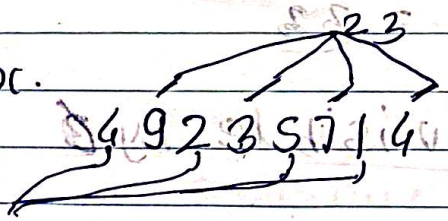
- sum of all digit & divide 9

Ex. 3714

8] Divisible by 11

A no is divisible by 11 is the difference of sum of its digits at odd places & sum of its digits at even places is either 0 or divisible by 11

Ex.



- difference of 22 & 12 is 11 is divisible by 11

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★ Add no of digits

① $1+2+3+4+\cdots+n = \frac{n(n+1)}{2}$

② $1^2+2^2+3^2+4^2+\cdots+n^2 = \frac{n(n+1)(2n+1)}{6}$

③ $1^3+2^3+3^3+4^3+\cdots+n^3 = \left\{\frac{n(n+1)}{2}\right\}^2$

★ Multiplication

Q. 6894×99

$= 6894 \times (100-1)$
$= 6894 \times 100 - 6894$
$= 689400 - 6894$
$= 61506$

Q. ~~829×9~~ $596 \times 596 - 104 \times 104$
$= 596^2 - 104^2$
$= (596-104)(596-104)$
$= 700 \times 492$
$= 344400$

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Q. $213 \times 213 + 187 \times 187$

$= 213^2 + 187^2$

~~$= (213 + 187$~~

$= \frac{1}{2}\left[(213+187)^2 + (213-187)^2\right]$

$= \frac{1}{2}\left[(400)^2 - (26)^2\right]$

$= \frac{1}{2}\left[160000 + 676\right]$

$= \frac{1}{2} \times \overset{80338}{\cancel{160676}}$

$= 80338$

Q $\dfrac{(637+478)^2 - (637-478)^2}{637 \times 478}$

$= \dfrac{(a+b)^2 - (a-b)^2}{ab}$

$= \dfrac{a^2 + b^2 - 2ab - a^2 + b^2 + 2ab}{ab}$

$= \dfrac{4ab}{ab} = 4$

3.2 → Point

3.2 → Multiplication

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Q $1^2 - 2^2 + 3^2 - 4^2 + \dots + 9^2 - 10^2 = ?$

$= (1+2)(1-2) + (3+4)(3-4) + (5+6)(5-6) + (7+8)(7-8) + (9+10)(9-10)$

$= -3 - 7 - 11 - 15 - 19$

$= -(3 + 7 + 11 + 15 + 19)$

$= -55$

Q $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \frac{1}{10.13} + \frac{1}{13.16}$

$= \frac{1}{3}\left(\frac{1}{1} - \frac{1}{4}\right) + \frac{1}{3}\left(\frac{1}{4} - \frac{1}{7}\right) + \frac{1}{3}\left(\frac{1}{7} - \frac{1}{10}\right) + \frac{1}{3}\left(\frac{1}{10} - \frac{1}{13}\right) + \frac{1}{3}\left(\frac{1}{13} - \frac{1}{16}\right)$

$= \frac{1}{3}\left[1 - \frac{1}{4} + \frac{1}{4} - \frac{1}{7} + \frac{1}{7} - \frac{1}{10} + \frac{1}{10} - \frac{1}{13} + \frac{1}{13} - \frac{1}{16}\right]$

$= \frac{1}{3}\left[1 - \frac{1}{16}\right]$

$= \frac{1}{3} \times \frac{15}{16} = \frac{5}{16}$

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Q 658 x 539 x 476 x 312

★ HCF & LCM

Q 10, 14

2 | 10 14

5 7

(2 common divisibles, both divisible by 2)

HCF = 2

LCM = 10 = 2 x 5
14 = 2 x 7
(2 common write one time and other also)

LCM = 2 x 7 x 5

= 70

Q 27, 63, 72

3 | 27, 63, 72

3 | 9 21 24

3 7 8

HCF = 3

LCM = 3 x 9 x 21 x 24

= 6

Product of no = HCF x LCM

56
3
168
3
504

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HCF = 3 x 3 = 9
LCM = 3 x 3 x 3 x 7 x 8
= 1512

Q HCF of 2 no is 7 & theere are LCM is 210 if of the no is 35 then find the other

let 2nd no x

35 x x = 7 x 210 , 42
x = ~~1470~~ 7 x 210
~~35~~ 35 5,
= 42

★ Co-prime

two no called coprime if one no is prime to each other.

i.e. HCF of 2 no is 1

Ex. 6, 11

1 | 6 11
6 11

6 & 11 are coprime.

HCF = HCF of numerators / LCM of denominators

LCM = LCM of numerators / HCF of denominators

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Q Find LCM & HCF 2/3, 8/9, 10/27 & 16/81

~~HCF = 2 + 8 + 10 + 16~~

HCF = HCF of 2, 8, 10, 16 / LCM of 3, 9, 27, 81

2 | 2, 8, 10, 16
1 4 5 8

3 | 3, 9, 27, 81
3 | 1, 3, 9, 27
3 | 1, 1, 3, 9
3 | 1, 1, 1, 3
1, 1, 1, 1

HCF = 2
LCM = 81

HCF = 2/81

LCM = ~~LCM of 2, 8, 10, 16~~
HCM of 3, 9, 27, 81

2 | 2, 8, 10, 16
4 | 1, 4, 9, 8
1, 1, 5, 2

3 | 3, 9, 27, 81
1, 3, 9, 27

= 2 x 4 x 5 x 2
= 80

LCM = 80/3

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Q Find the lagest no exactly divide 513, 783, & 1107

	513	783	1107
3	171	261	369
3	57	87	123
	19	29	41

HCF = 3 x 3 x 3
= 27

Q find the smallest no exatly divide by each one of the no 12, 15, 20, 27

	12	15	20	27
3	4	5	20	9
4	1	5	5	9
5	1	1	1	9
	1	1	1	9

= 3 x 4 x 5 x 9
= 540

Q 35, 823, 555, 400

	35	823	555	400
5	7	823	111	80
3	7	823	37	80
	7	823	37	80

HCF = 15
LCM = 15986750

☆ Decimal fraction.

$\frac{1}{10} = 0.1$ $\quad\quad$ $\frac{1}{1000} = 0.001$

conversation of decimal no to vulgar fraction

$$0.5 = \frac{5}{10} = \frac{1}{2}$$

$$0.56 = \frac{56}{100} = \frac{14}{25}$$

Q $345.9 + 34.59 + 3.459 + 0.3459$

$$\begin{array}{r} {}^{1}{}^{2}{}^{1} \\ 345.91 \\ 34.591 \\ 3.459 \\ 0.3459 \\ \hline 384.2949 \end{array}$$

Q $1.11 \times 2.1 \times 3.2$

$$\begin{array}{r} 1.11 \\ 2.10 \\ \hline 1110 \\ 22200 \\ \hline 23310 \end{array} \qquad \begin{array}{r} 233.10 \\ 3.2 \\ \hline 666200 \\ 6993000 \\ \hline 7.459200 \end{array}$$

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Q Divide 35 by 0.007

$$= \frac{35000}{7} = 5000$$

★ Recurring Decimal

Ex. $0.\bar{3}$ ($\bar{3} = 0.333\ldots$)

$$= \frac{3}{9} = \frac{1}{3}$$

(9 is one bar = 10-1
2 bar = 100-1)

Ex - $0.2\bar{6}$ (bar nahi be use -ve • kro or niche extra 0 de do)

$$= \frac{26-2}{90} = \frac{24}{90}$$

Ex $3.46\bar{7} = \frac{3+467-46}{900} = 3\frac{421}{900}$

$=$

$a^3+b^3+c^3-3abc = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$

$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$

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Q $\frac{3}{8}, \frac{7}{12}, \frac{2}{3}, \frac{16}{25}, \frac{14}{15}$

$= 0.37, 0.58, 0.67, 0.64, 0.93$

$= 0.37, 0.58, 0.64, 0.67, 0.93$

★ Simplification

Q $\frac{(a-b)^3+(b-c)^3+(c-a)^3}{6(a-b)(b-c)(c-a)}$

Let $x = a-b$

$y = b-c$

$z = c-a$

$x+y+z = 0$

$= \frac{x^3+y^3+z^3}{6xyz} = \frac{3xyz}{6xyz}\left(\frac{x^3+y^3+z^3 = 3xyz}{x^3+y^3+z^3 = 3xyz}\right)$

$= \frac{3xyz}{6xyz} = \frac{1}{2}$

Q If $a+b+c = 13$ & $a^2+b^2+c^2 = 69$ then $(ab+bc+ca) = ?$

$(a+b+c)^2 = a^2+b^2+c^2+2(ab+bc+ca)$

$(13)^2 = (69) + 2(ab+bc+ca)$

$169-69 = 2(ab+bc+ca)$

$\frac{100}{2} = ab+bc+ca$

$50 = ab+bc+ca$

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So = ab + bc + ca

Q If $x + \cfrac{1}{1 + \cfrac{1}{3 + \cfrac{1}{4}}} = 2$ then $x = ?$

$$x + \frac{1}{1 + \frac{1}{\frac{13}{4}}} = 2$$

$$x + \frac{1}{1 + \frac{4}{13}} = 2$$

$$x + \frac{1}{\frac{17}{13}} = 2$$

$$x + \frac{13}{17} = 2$$

$$x = \frac{13}{17} - 2$$

$$x = \frac{17 - 13}{17} = \frac{4}{17}$$

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Q If $\frac{a}{b} = \frac{2}{3}$ then find $\frac{3a+5b}{3a-5b}$

$$\frac{3\frac{2}{3}+5\frac{2}{3}}{3\frac{2}{3}+5\frac{2}{3}} = \frac{\frac{18}{3}+\frac{30}{3}}{\frac{18}{3}-\frac{30}{3}} = \frac{1}{3}\left[\frac{18+30}{18-30}\right]$$

$\frac{a}{b} = \frac{2}{3}$ $\quad 3a = 2b$

$$\frac{3a+5b}{3a-5b} = \frac{3a}{3a} = \frac{3\frac{2}{3}}{3\frac{2}{3}}$$

$$\frac{3(b)+5(3)}{b} \quad \frac{2b+5b}{2b-5b}$$

$$= \frac{7b}{-3b} \quad [\because b \neq 0]$$

$$= -\frac{7}{3}$$

Q Find $\left(1-\frac{1}{3}\right)\left(1-\frac{1}{4}\right)\left(1-\frac{1}{5}\right)\cdots\left(1-\frac{1}{99}\right)\left(1-\frac{1}{100}\right)$

$$= \frac{2}{3} \times \frac{3}{4} \times \frac{4}{5} \times \cdots \times \frac{98}{99} \times \frac{99}{100}$$

$$= \frac{2}{3} \times \frac{99}{100}$$

$$= \frac{2}{100} = \frac{1}{50}$$

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Q $\frac{1}{5} + 999\frac{494}{495} \times 99 = ?$

$= \frac{1}{5} + \left(999 + \frac{494}{495}\right) \times 99$

$= \frac{1}{5} + \left((1000 - 1) + \frac{494}{495}\right) \times 99$

$= \frac{1}{5} + \left(99000 - 99 + \frac{494}{495 \times 5}\right) \times 99$

$= \frac{1}{5} + 99000 - 99 + \frac{494}{5}$

$= \frac{495}{5} + 99000 - 99 = 99 + 99000 - 99$

$= 99000$

Q $\cfrac{1}{1 + \cfrac{\frac{2}{3}}{1 + \frac{2}{3} + \cfrac{\frac{8}{9}}{1 - \frac{2}{3}}}}$

$= \cfrac{1}{1 + \cfrac{2/3}{1 + \frac{2}{3} + \cfrac{8/9}{1/3}}}$

$= \cfrac{1}{1 + \cfrac{2/3}{1 + \frac{2}{3} + \frac{8}{3}}}$

$= \cfrac{1}{1 + \cfrac{2/3}{1 + \frac{10}{3}}}$

$= \cfrac{1}{1 + \cfrac{2/3}{\frac{13}{3}}}$

$= \cfrac{1}{1 + \frac{2}{13}}$

$= \frac{13}{15}$

$$= \frac{1}{1 + \frac{2/3}{1 + \frac{2}{3} + \frac{8}{93} \times 31}}$$

$$= \frac{1}{1 + \frac{2/3}{1 + \frac{10}{3}}} = \frac{1}{1 + \frac{2/3}{\frac{13}{3}}}$$

$$= \frac{1}{1 + \frac{2}{3} \times \frac{3}{13}} = \frac{1}{\frac{15}{13}} = \frac{13}{15}$$

Q If $\frac{3}{7}$ of $\left(\frac{2}{5}\right)$ of a number is 198, find no.

$$= \frac{3}{7} \times \frac{2}{5} \quad a = 198$$

$$\frac{2}{5} a = \frac{198 \times 3}{7}$$

$$= \frac{6}{35} \quad a = 198$$

$$= \quad a = \frac{198 \times 35}{6}$$

$$= 33 \times 35 = 1155$$

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Q A boy was ask ~~to~~ to multiply a no 25. bi insted multiply by no 52 & got the ans 324 more than the correct ans what was the no

$25x = 52 + 324$

$= 376$

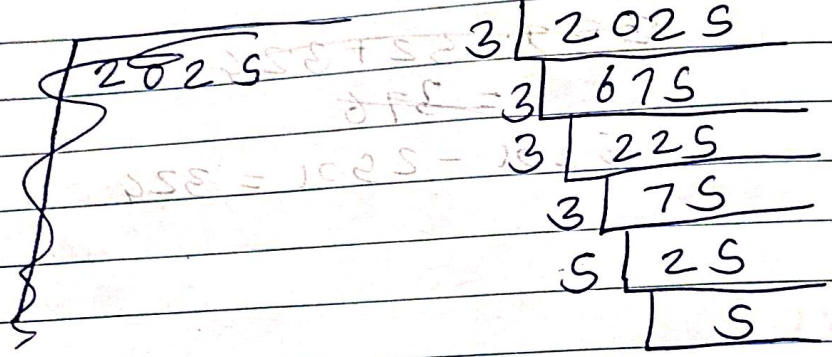
$52x - 25x = 324$

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☆ Square Root & Cube root:

Ex. $\sqrt{2025}$



$\sqrt{3 \times 3 \times 3 \times 3 \times 5 \times 5}$ (check double value is present then take one time)

$= 3 \times 3 \times 5$

$= 45$

Ex.

$\sqrt{1734489}$

```
      1317
   1 | 1734489
     | 1
  23 | 73
     | 69
 261 | 444
     | 261
2627 | 18389
     | 18389
```

(seperet 2 digits & check Fist divisible by)

(you can take 1st digit then multiply by 2 & checke divisible)

(in uppre 13 multiply 2 & and ~~add on~~ divisible & find)

Q $\sqrt{248+\sqrt{51+\sqrt{169}}}$

$= \sqrt{248\sqrt{51+13}}$

$= \sqrt{248\sqrt{64}} \quad = \sqrt{248+8} \quad = \sqrt{256}$

$= \underline{16}$

Q If $\sqrt{1369} = 37$ then find the value of $\sqrt{13.69} + \sqrt{0.1369} + \sqrt{0.001369} + \sqrt{0.0000\,1369}$

$= \sqrt{\frac{1369}{100}} + \sqrt{\frac{1369}{10000}} + \sqrt{\frac{1369}{1000000}} + \sqrt{\frac{1369}{100000000}}$

$= \frac{37}{10} + \frac{37}{100} + \frac{37}{1000} + \frac{37}{10000}$

$= 3.7 + 0.37 + 0.037 + 0.0037$

$= \underline{4.1107}$

Q $\sqrt[3]{\frac{7}{875}}$

$= \sqrt[3]{\frac{1}{125}} \quad = \frac{1}{5}$

5x5x5

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Q $\sqrt[3]{\sqrt{0.000064}}$

$= \sqrt{\sqrt{\frac{64}{1000000}}}$

$= \sqrt{\frac{8}{1000}} = \underline{\underline{0.2}}$

Q $\frac{\sqrt{x+4} + \sqrt{x-10}}{\sqrt{x+4} - \sqrt{x-10}} = \frac{5}{2}$, $x = ?$

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☆ Average ☆

Formula:-

$$\frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

A man ~~foo~~ covers d at x km/hr & returns back to the starting point y km/hr the average speed during whole journey is $\frac{2xy}{x+y}$ km/hr

Q Find first 100 natural number avg

$$\frac{n(n+1)}{2}$$

$$= \frac{100(100+1)}{2} = \frac{101}{2} = 50.5$$

Q Fist 5 multiples of 3 avg

$$\frac{3+6+9+12+15}{5} = \frac{45}{5} = 9$$

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Q The avg of 5 no is 27 if one no is excluded the avg become 25 find excluded no.

Avg of 5 no = 27 ~~x5 = 135~~

sum of 5 no = 27 x 5 = 135

$$\frac{135 - x}{4} = 25$$

$$135 - x = 100$$

$$x = 100 - 135$$

$$x = 35$$

3 years ago the avg age of a family of 5 members is 17 years. A baby having being bond, The avg is of family is same. Find the present age of the baby.

~~x = baby age~~ x = baby age

~~n = 6~~

3 y age sum of age = 17 x 5 = 85 years

Today, after 3 years = 85 + 3 x 5 = 85 + 15 = 100 years

$$\frac{100 + x}{6} = 17$$

$$x = 102 - 100$$

$$= 2$$

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Q A ship sell out to a 15 km/hr & sells back to a starting point of a 10 km/hr. Find avg rate of selling of ship.

$x = 15 \qquad y = 10$

$$\frac{2xy}{x+y}$$

$$= \frac{2 \times 15 \times 10}{15+10} = \frac{300}{25} = \underline{\underline{12}}$$